Ahmed Dorra

AI Engineer — LLM Applications, Machine Learning & Python Backends

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SUMMARY

AI engineer who builds AI products end to end in Python: LLM applications (RAG, agents and their evaluation), model fine-tuning with PyTorch and Hugging Face, and the FastAPI services, tests and monitoring that run them. On a 12,730-product shopping assistant, raised right-product-ranked-first from 23% to 43% and agent task success from 7 to 12 of 12 tasks. Published researcher (LoRA fine-tuning of Stable Diffusion; quantum machine learning), Computer Science teaching assistant, and native Arabic speaker with professional English.

SKILLS

- **LLM Applications:** LangGraph, LangChain, retrieval-augmented generation (RAG), hybrid search (PostgreSQL full-text + pgvector), cross-encoder reranking, agents and tool calling, conversational memory, prompt engineering, structured output (Pydantic), guardrails, Groq, Cohere, Chroma
- **Machine Learning:** PyTorch, Hugging Face Transformers, Diffusers, LoRA fine-tuning, scikit-learn, XGBoost, pandas, NumPy, spaCy, NLTK
- **Evaluation & MLOps:** labelled test sets with confidence intervals, code-graded evaluation, LLM-as-a-judge, RAGAS, Langfuse (tracing, prompt management), MLflow, DVC, Prometheus, Grafana
- **Backend & Deployment:** Python, FastAPI (REST APIs), SQL, PostgreSQL, SQLAlchemy, Alembic, Git, Docker, Docker Compose, GitHub Actions (CI), pytest, Nginx, Gradio, Hugging Face Spaces

PROJECTS

Shopping Assistant — LLM agent and RAG service behind a FastAPI API GitHub

Python · FastAPI · LangGraph · PostgreSQL · pgvector · Cohere · Groq · Langfuse · MLflow · Prometheus · Grafana · Docker

- Built a LangGraph agent, served through a FastAPI REST API on PostgreSQL, that searches a 12,730-product catalogue with hybrid retrieval (PostgreSQL full-text + pgvector HNSW, Reciprocal Rank Fusion, Cohere reranking) and remembers each customer's lasting preferences across conversations.
- Measured retrieval stage by stage on 127 labelled queries with 95% confidence intervals: hybrid top-10 recall 72% vs 56% keyword and 41% vector; reranking lifted right-product-first from 23% to 43% (40-query subset); an HNSW index cut median vector search from 35 ms to 3.1 ms.
- Designed a code-graded agent evaluation (12 tasks × 3 runs, logged to Langfuse and MLflow): prompt v2 passed 12 of 12 tasks vs 7, with 29% fewer paid rerank calls; moving price limits inside search lifted on-target results from 70% to 100%.
- Added input and memory-write guardrails (0 of 21 legitimate facts refused), Langfuse tracing and prompt versioning, Prometheus/Grafana monitoring, 104 automated tests and GitHub Actions CI.

Research Assistant — Agentic RAG over arXiv and the web Live demo · GitHub

Python · LangGraph · LangChain · Chroma · BM25 · FastAPI · Gradio · Docker · Hugging Face Spaces

- Designed a self-correcting LangGraph workflow: query rewriting, hybrid retrieval, relevance grading, cited generation and a reflection step with a bounded retry, using structured outputs for each decision.
- Combined dense (Chroma) and keyword (BM25) retrieval with cross-encoder reranking; a provider factory switches between Groq, Hugging Face, Anthropic and OpenAI with one setting; served through FastAPI and Gradio and deployed on Hugging Face Spaces.

VISIONEER — Text-to-floor-plan generation with AR visualisation (2024) Model · Paper

Stable Diffusion v1.5 · Hugging Face Diffusers · LoRA · Android · ARCore

- Fine-tuned Stable Diffusion v1.5 with LoRA (Hugging Face Diffusers) on ~12k captioned floor plans to generate furnished 2D layouts from text prompts; a teammate trained the companion schematic model, and both were served to an Android AR app. Published in Machine Graphics & Vision (2025).

EXPERIENCE

Teaching Assistant, Computer Science — Culture and Science City · Giza 03/2026 – Present

- Teach Advanced Operating Systems to 700+ students: processes and threads, CPU scheduling, synchronization and deadlocks, memory management and virtual memory, and file systems.
- Teach a starter MLOps course to about 500 students, modelled on DataTalks.Club's ML Zoomcamp: data preprocessing and cleaning, a linear model trained on a customer-churn dataset, then an industry-standard project layout, experiment tracking with MLflow, and serving the model with FastAPI and Docker.
- Assisted the supervision of two 2025/26 graduation projects: an AI system for social-media trend and emotion analysis, and diabetes detection with machine learning.
- Designed and led 50 Compiler Theory lab sessions for 600+ students; co-taught neural-network and CNN sessions; set Distributed Computing quizzes; led Linear Algebra, Calculus and Information Retrieval tutorials.

Military Service (mandatory) — Egypt 01/2025 – 01/2026

Quantum Research Intern — QUAE (QWorld UAE chapter) · Remote 12/2023 – 03/2024

- Built a quantum machine-learning pipeline for EEG classification: a Quantum Variational Autoencoder for self-supervised denoising, followed by LDA for cognitive-task classification in brain–computer interfaces.
- Evaluated generalisation and computational-complexity trade-offs on noisy neurological time series; results published at the Neurogenomics 2024 Conference.

PUBLICATIONS

- **Fine-tuning Stable Diffusion for Generating 2D Floor Plans Using Prompt Templates** — Machine Graphics & Vision, 34(3), 77–95 (09/2025) · mgv.sggw.edu.pl/article/view/10294
- **Quantum Denoising in the Realm of Brain-Computer Interfaces: A Preliminary Study** — Neurogenomics 2024 Conference (07/2024) · hal.science/hal-04501908

ORCID: orcid.org/0009-0001-4161-6026

EDUCATION

Bachelor of Science in Computer Science — Culture and Science City · 6th of October, Giza 10/2020 – 06/2024

Relevant coursework: Machine Learning & AI, Software Engineering, Data Structures & Algorithms.

Diploma in Quantum Computing — QWorld · Remote 10/2023 – 07/2024

Quantum algorithms, quantum machine learning and quantum information; QBronze, QSilver and QNickel certifications.

TRAINING

- MLOps Practitioner Course — MLOps MENA Community (2026)
- Mini-RAG: production RAG systems — Abu Bakr Soliman (2026)
- Hugging Face LLM Course (in progress)
- Generative AI with Large Language Models — DeepLearning.AI
- Natural Language Processing Specialization — DeepLearning.AI
- Deep Learning Specialization — DeepLearning.AI
- Machine Learning Specialization — DeepLearning.AI & Stanford
- Applied Text Mining in Python — University of Michigan
- Vector Databases (Pinecone) — 365 Data Science

LANGUAGES

Arabic (native) · English (professional working proficiency)